Day 2 of 15 · Student Activity Sheets

Day 2 Activities — Descriptive Statistics in Excel

TESDA Data Analytics Level III · Batch DAL3-2026

Name: _____ Date: _____ Workstation #: _____

Files you need (in `Day02/assets/`): `Day02_Descriptive_Stats_Template.xlsx` (main working file) and `Day02_Branch_Sales.xlsx` (raw data, keep it untouched).

Before you begin:

1. Copy the files into `DAL3/Day02/`.
2. Open the template and **Save As** `Day02_<Surname>.xlsx`.
3. Look at the sheets: **Stats**, **Frequency**, **Outliers**, **Daily_Sales**, **Daily_Transactions**.

Activity	Title	Mode	Time	Points
2.0	Warm-up: Compute by Hand	Individual	20 min	10
2.1	Descriptive Statistics with Excel Functions	Individual	50 min	30
2.2	Cross-check with the Analysis ToolPak	Individual	25 min	15
2.3	Frequency Distribution and Histogram	Individual	40 min	20
2.4	Flag the Outliers	Pair	25 min	10
2.5	Insight Summary for Aling Nena	Group of 3–4	30 min	15
	Total		**≈ 3 hrs 10 min**	**100**

Activity 2.0 — Warm-up: Compute by Hand (10 pts)

Marikina's number of transactions for 8 days: **120, 115, 130, 120, 125, 140, 120, 30**

Show your solution in the space below. Use a calculator, not Excel.

Measure	Your answer
Sorted data	
Mean	
Median	
Mode	
Range	

Measure	Your answer
Which value looks like an outlier? Which average best represents a "typical day"? Why?	

Activity 2.1 — Descriptive Statistics with Excel Functions (30 pts)

On the **Stats** sheet, enter a formula in every yellow cell (rows 4–20) for all four branches.

Data ranges:

Branch	Daily_Sales range	Daily_Transactions range
Quezon City	`Daily_Sales!C2:C91`	`Daily_Transactions!C2:C91`
Marikina	`Daily_Sales!D2:D91`	`Daily_Transactions!D2:D91`
Pasig	`Daily_Sales!E2:E91`	`Daily_Transactions!E2:E91`
Caloocan	`Daily_Sales!F2:F91`	`Daily_Transactions!F2:F91`

Steps

1. In **B4** (Count, Quezon City) type `=COUNT(Daily_Sales!C2:C91)` → it should give **90**.
2. Complete column B (Quezon City), using the Excel Function Summary table in your handout:
   - Mean `=AVERAGE(...)`, Median `=MEDIAN(...)`, Mode `=MODE.SNGL(Daily_Transactions!C2:C91)`
   - Min, Max, Range `=B9-B8`, Sample Variance `=VAR.S(...)`, Sample Std Deviation `=STDEV.S(...)`
   - CV (%) `=B12/B5*100`, Q1 `=QUARTILE.INC(...,1)`, Q3 `=QUARTILE.INC(...,3)`, IQR `=B15-B14`
   - Lower Fence `=B14-1.5*B16`, Upper Fence `=B15+1.5*B16`
   - No. of Outliers `=COUNTIF(range,"<"&B17)+COUNTIF(range,">"&B18)`, Skewness `=SKEW(...)`
3. Fill columns C, D and E: select **B4:B20** → **Ctrl + C** → select **C4:E20** → **Ctrl + V**. The ranges are relative references, so they shift to the next branch automatically (e.g. `C2:C91` becomes `D2:D91`). Click one cell in column E and confirm it points to `F2:F91`.
4. Format the peso values with a thousands separator: **Home** → **Comma Style**.

Questions

a) Which branch has the **highest mean** daily sales? _____ The **least consistent** sales (highest CV)? _____

b) For which branch is the mean **much larger** than the median? What does that tell you about its shape?

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c) Why do we use `STDEV.S` and not `STDEV.P` here?

Activity 2.2 — Cross-check with the Analysis ToolPak (15 pts)

(Integrated: Apply quality standards.)

1. Enable the ToolPak if needed: **File → Options → Add-ins → Manage: Excel Add-ins → Go → ✓ Analysis ToolPak → OK**.
2. **Data → Data Analysis → Descriptive Statistics → OK**.
3. Input Range: `Daily_Sales!$C$1:$F$91` · Grouped By: **Columns** · ✓ **Labels in first row**.
4. Output: **New Worksheet Ply**, named `ToolPak` · ✓ **Summary statistics → OK**.
5. Widen the columns (select them, then double-click a column border).
6. Fill in the comparison:

Branch	Mean: function (2.1)	Mean: ToolPak	Std Dev: function	Std Dev: ToolPak	Match? ✓/✗
Quezon City					
Marikina					
Pasig					
Caloocan					

d) The ToolPak shows **Mode = #N/A** for daily sales. Why? Why did we compute the mode from **Daily_Transactions** instead?

Activity 2.3 — Frequency Distribution and Histogram (20 pts)

On the **Frequency** sheet (all branches combined, n = 360):

1. **D4**, frequency: `=COUNTIFS(Daily_Sales!$C$2:$F$91,">="&A4,Daily_Sales!$C$2:$F$91,"<="&B4)`. Fill it down to D14.
2. **D15** total: `=SUM(D4:D14)`. It must equal **360**. If it doesn't, check your formula; this is your quality check.
3. **E4**, relative frequency %: `=D4/$D$15*100` → fill down.
4. **F4** cumulative: `=D4` ; **F5**: `=F4+D5` → fill down. F14 must equal 360.
5. **Make the histogram:** select **C3:D14** → **Insert** → **Column Chart (2-D Clustered Column)**.
   - Right-click a bar → **Format Data Series** → **Gap Width = 5%**.
   - Chart title: *Histogram of Daily Sales (all branches)*. Axis titles: *Daily sales (PHP)* and *Number of days*.
6. *(Optional)* Try the built-in histogram: select `Daily_Sales!C2:F91` → **Insert** → **Statistic Chart** → **Histogram**.

e) Which class has the highest frequency? _____ Is the distribution symmetric, right-skewed, or left-skewed? _____

f) Look at the bars far to the right and far to the left. What kinds of days might they be?

Activity 2.4 — Flag the Outliers (Pair, 10 pts)

On the **Outliers** sheet (Pasig branch):

1. **B2** = `=Stats!D17` (lower fence), **B3** = `=Stats!D18` (upper fence).
2. **C6**: `=IF(OR(B6<$B$2,B6>$B$3),"OUTLIER","")` → fill down to C95.
3. Highlight outliers: select **A6:C95** → **Home** → **Conditional Formatting** → **New Rule** → **Use a formula** → `=$C6="OUTLIER"` → Format → Fill light red → OK.
4. List the outlier dates below, then discuss with your partner whether each is a **data error** or a **real event**. Hint: check the day of the month and the weekday.

Date	Pasig Sales (₱)	Possible explanation	Keep / Investigate

Activity 2.5 — Insight Summary for Aling Nena (Group, 15 pts)

(Integrated: Lead workplace communication; Facilitate entrepreneurial skills for MSMEs.)
As a team, write **5 sentences** in the yellow area on the **Stats** sheet (rows 23–27), using **What** → **So what** → **Now what**:

1. Which branch sells the most and the least per day (use the mean or median)?
2. Which branch is the most and the least consistent (use the CV)?
3. What do the outliers tell us (paydays? fiesta? flood?)
4. What is the shape of the sales distribution, and which "average" should the owner use?
5. One practical recommendation (stocking, staffing, promos).

The reporter presents in 2 minutes. **Rubric:** accuracy of numbers (5), clarity for a non-technical owner (5), quality of the recommendation (5).

Daily Reflection Log

1. The statistic I understand best today, and why:

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2. Something I want to practice more:

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3. How could a small business near me use the median or the standard deviation?

Submit: Day02_<Surname>.xlsx (with Stats, ToolPak, Frequency + histogram, Outliers sheets) and this sheet.